

Cross-level steam load smoothing and optimization in industrial parks using data-driven approaches

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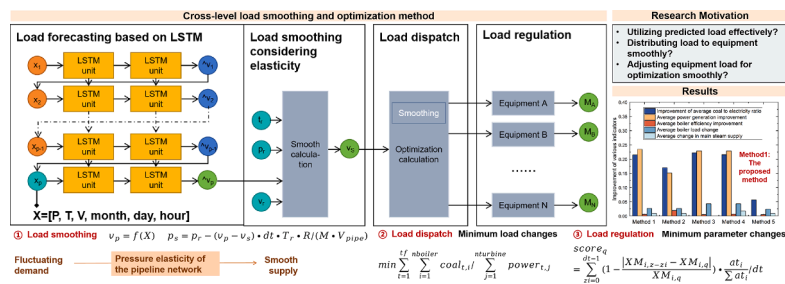
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HIGHLIGHTS

- A cross-level method for load smoothing regulation of industrial parks is proposed.
- Heat network elasticity is presented as a buffer between demand and supply.
- A method for utilizing load forecasting is developed.
- Constraints for minimal changes in load and operational parameters are established.
- Balance between efficiency and equipment load stability is achieved.

GRAPHICAL ABSTRACT



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ABSTRACT

This study focuses on the integrated energy production system in industrial parks, addressing the problem of stable load dispatch of equipment under demand fluctuations. A cross-level method for steam load smoothing and optimization is proposed, aiming to achieve stable production and optimal economic performance through three levels of integration: load forecasting, load dispatch, and load regulation. Unlike traditional methods that directly use load forecasting values, heat network elasticity is presented as a buffer between demand and supply. Constraints for minimal changes in equipment load and operational parameters are established for smooth regulation. Industrial cases demonstrate that the load forecasting model has mean absolute percentage errors of 2.44% and 1.68% for medium-pressure and low-pressure steam, respectively, meeting accuracy requirements. The modified supply-side load smoothness is effectively improved by considering heat network elasticity. The method increases boiler efficiency by 1.92%, reducing average coal consumption by 0.92 t/h. Compared to manual operation, the proposed model leads to an average increase of 5.69 MW in power generation and an average reduction of 10.81% in coal-to-electricity ratio. This study verifies the importance of smooth integration across different levels and analyzes the effective response of the proposed method to the uncertainty in load forecasting. The method demonstrates the enormous potential of data-driven methods in achieving safe, economical, and sustainable production in industrial parks.

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Nomenclature			
Acronyms			
CNN	convolutional neural networks	$P_{t,j}$	generating capacity of steam turbine j at time t
GA	genetic algorithm	$\mathbf{p}$	parameters about the pressure
GP	Gaussian process	$\mathbf{p}_r$	actual pressure of the pipe
IP	industrial parks	$\mathbf{p}_s$	pressure of pipe after smooth
LS-SVM	least square support vector machine	R	gas constant
LSTM	long short-term memory	t_b	past time period
MAE	mean absolute error	t_f	time length of predicted load
MAPE	mean absolute percentage error	$U_{bs,i}$	upper limit of the steam production of boiler i
MSE	mean square error	$U_{coal,i}$	upper limit of coal consumption of boiler i
NSI	non-smoothness index	$U_{dbs,i}$	upper limit of ramping
TDPR	temperature-decreased pressure reducer	$U_{p,j}$	upper limit of power generation of turbine j
PSO	particle swarm optimization	$U_{tin,j}$	upper limit of the steam inlet of turbine j
RMSE	root mean square error	$U_{tout0,j}$	upper limit of extraction flow of turbine j
SI	smoothness index	$U_{tout1,j}$	upper limit of exhaust flow of turbine j
Symbols		U_{vs}	upper limit of pipe quantity
d_t	time interval	V_{pipe}	capacity of the pipe
$E_{b,t,i}$	efficiency of boiler i at time t	$\mathbf{v}$	parameters about the quantity
$H_{b,t,i}$	enthalpy value of boiler steam i at time t	$v_{atout,t,q}$	steam output of TDPR q at time t
$H_{coal,t,i}$	enthalpy value of feed coal of boiler i at time t	$v_{bs,t,i}$	steam production of boiler i at time t
$L_{bs,i}$	lower limit of the steam production of boiler i	$v_{cin,t,z}$	steam inlet of compressor z at time t
$L_{coal,i}$	lower limit of coal consumption of boiler i	$v_{coal,t,i}$	coal consumption of boiler i at time t
$L_{dbs,i}$	lower limit of ramping	v_p	predicted load
$L_{p,j}$	lower limit of power generation of turbine j	$v_{tin,t,j}$	steam inlet of turbine j at time t
$L_{tin,j}$	lower limit of the steam inlet of turbine j	$v_{tout0,t,j}$	extraction flow of turbine j at time t
$L_{tout0,j}$	lower limit of extraction flow of turbine j	$v_{tout1,t,j}$	exhaust flow of turbine j at time t
$L_{tout1,j}$	lower limit of exhaust flow of turbine j	$\mathbf{v}_r$	actual quantity of the pipe
L_{vs}	lower limit of pipe quantity	$\mathbf{v}_s$	quantity of pipe after smooth
M	molar mass	$\mathbf{v}_{s,0}$	initial value of $\mathbf{v}_s$
n_{aprd0}	total number of low-pressure TDPRs	$v_{sup0,s,t}, v_{sup1,s,t}$	total amount of different steam supply after load
n_{aprd1}	total number of medium-pressure TDPRs	$\mathbf{X}$	input parameter
n_{boiler}	total number of boilers	$\mathbf{x}_d$	parameters about the day
$n_{compressor}$	total number of compressors	$\mathbf{x}_h$	parameters about the hour
n_{change}	number of boilers that require load adjustments	$\mathbf{x}_m$	parameters about the month
$n_{turbine}$	total number of turbines	θ	parameters about the temperature
		θ_r	actual temperature of the pipe

1. Introduction

1.1. Background

Global climate change constitutes the most significant environmental challenge at present [1,2], and promoting the development of green industries is viewed as one of the key strategies for achieving a sustainable transition to a low-carbon society [3]. Industrial parks (IPs), in this context, hold a substantial influence on this transition [4,5]. As per statistical data, IPs are responsible for nearly 69% of society's total energy consumption, and their carbon emissions constitute approximately 31% of the nation's total emissions [6]. Strengthening the energy management and regulation within IPs, optimizing energy consumption, and improving energy utilization efficiency are critically important for the attainment of carbon neutrality.

IPs typically encompass a wide range of energy production, transmission, and consumption activities, with improving the energy efficiency of the energy production system being of paramount importance [7]. As depicted in Fig. 1, a typical IP is shown. Within this IP, the integrated energy production system is responsible for energy conversion, supplying downstream users with energy such as steam and compressed air. These downstream users are primarily printing and dyeing and textile enterprises. Steam plays a pivotal role in these enterprises [8]. Despite its small volume, its application range is extremely broad [9,10].

The demand of these steam users is influenced by a variety of factors such as raw material prices, market demand, and trade forms, leading to significant demand fluctuations. Although some large-scale steam users have a certain periodicity in their demand, a large number of users have small and irregular steam flows.

Therefore, for the integrated energy production system, the key decision is how to coordinate the load regulation of internal boilers, turbines, compressors, and other equipment to meet the fluctuating demand of downstream users, while maximizing energy efficiency and minimizing carbon emissions. This is essentially a load dispatch problem.

1.2. Literature review

Many researchers have studied on load dispatch problems, with the main methods including optimization-based, simulation-based, or priority-based methods [11]. Among these, optimization-based methods are the mainstream approach. For example, Boix et al. [12] proposed a method based on mixed integer linear optimization for water systems, and further explored energy optimization methods among multiple companies in IPs from a design perspective [13]. Cao et al. [14] adopted a distributed robust optimization algorithm to solve the microgrid optimization problem considering renewable energy. Lü et al. [15] proposed the genetic algorithm (GA) method to solve the unit

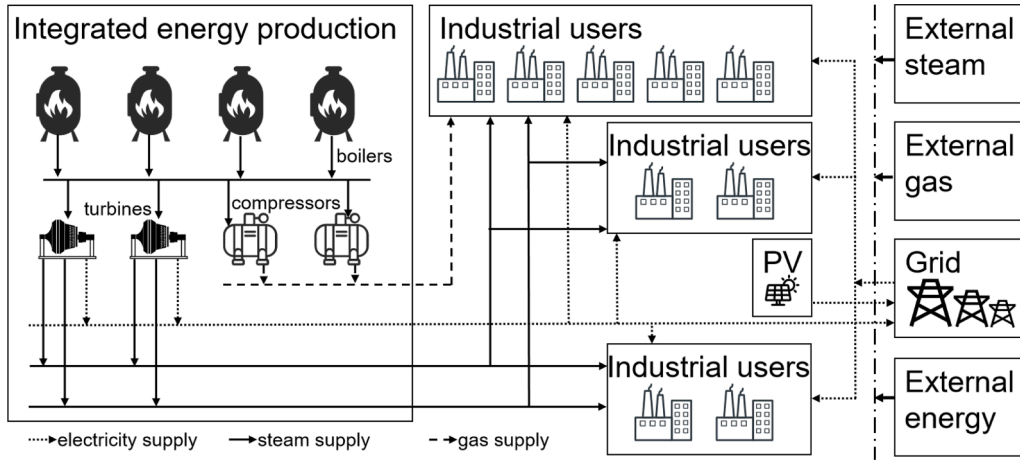


Fig. 1. A typical flow chart of the IP.

commitment scheduling problem. Ocampo et al. [16] constructed a two-stage unit optimization model based on day-ahead robust scenarios, using a method called block step particle swarm optimization (PSO) for solving. In addition, Ponciroli et al. [17] and Zhai et al. [18] improved the traditional heuristic methods in the solution process of the scheduling optimization problem. With the development of artificial intelligence algorithms, researchers such as Tiwari et al. [19] proposed a multi-stage solution method based on hybrid artificial intelligence for the load dispatch problem of traditional thermal power and renewable energy storage power systems. These methods, oriented towards economic objective optimization, can achieve benefit enhancement and provide model support for the operation scheduling of the integrated energy production system in IPs.

However, load dispatch merely provides the optimized load values for the internal equipment of the integrated energy production system. How to execute the load has become one of the key challenges in the operation scheduling of the IP. Different operation methods can achieve the same load value. Researchers like Zhu et al. [20] point out that the efficiency coefficients of equipment may vary depending on actual operation conditions. Zhou et al. [21] also demonstrated this point in their research on optimizing heat recovery systems.

Further research on load regulation is necessary. In practical implementation, the regulation mode of IPs' equipment is mainly based on human experience. In research, various methods have been explored in boiler regulation, including clustering, regression modeling, online monitoring, and pattern mining. Jia et al. [22] used pattern mining to establish a reasonable fuzzy partition matrix to optimize boiler operation, while Wang et al. [23] used the Gaussian process (GP) to simulate the relationship between NO_x emission characteristics and boiler operating parameters. Similarly, Kang et al. [24] established a least squares support vector machine (LS-SVM) model and used an adaptive GA to optimize the boiler combustion process. Tang et al. [25] developed a multi-objective optimization model based on a deep belief network. Blackburn et al. [26] used machine learning to dynamically model coal-fired power plants and determine adjusted control variables through PSO. The preceding studies have provided support for the optimization of equipment load regulation. Nevertheless, existing research using a single regression model may face difficulties in adapting to changes in working conditions. Moreover, optimization based on regression models may lead to extrapolation errors, resulting in load regulation schemes that deviate from reality. Furthermore, existing studies seldom consider the stability of operation parameter adjustment, further reducing feasibility.

Beyond the challenges of feasibility, traditional load dispatch also encounters the issue of equipment load fluctuation. The primary reason is that the traditional dispatch model for IPs needs to consider the energy demand of IP users as a constraint. It adjusts the equipment load to meet

the demand load, requiring supply and demand to balance at the same point in time. Wu et al. [27] pointed out that the analysis of industrial load demand has more pronounced randomness and volatility compared to the analysis of cooling and heating loads in the construction industry. The volatility of demand inevitably leads to fluctuations in the load regulation strategies of energy production system equipment, thereby reducing equipment efficiency and lifespan [28]. Therefore, improving the traditional supply-demand matching mode and reducing equipment load fluctuations is the key challenges faced by the integrated energy production system in IPs.

Currently, with the advancement of artificial intelligence technology, understanding the volatility of demand in advance through load forecasting methods has become one of the ways to improve the existing supply-demand matching mode. Kerkkänen et al. [29] has mentioned the importance of demand forecasting on the supply chain. Li et al. [30] proposed an AI process and potential architecture from load forecasting to control. However, due to the complexity brought about by factors such as user diversity, energy consumption patterns, seasonality, time, weather, and policy, predicting the energy demand in IPs remains a challenge [31]. It is difficult to use mechanistic models for industrial demand prediction. Therefore, data-driven load forecasting methods, including characteristic-based, time-series-based, and hybrid methods, have been widely adopted. Alkhayat et al. [32] specifically pointed out that hybrid methods are the most commonly used. Researchers have proposed various models for load forecasting using techniques such as artificial neural networks [33], LS-SVM [34], temporal convolutional networks [35], and improved long short-term memory (LSTM) [36]. Moreover, Ribeiro et al. [37] introduces a load forecasting framework based on wavelet neural networks. Liu et al. [38] proposed a clustering and dynamic recognition-based auto-reservoir neural network for power load forecasting. Some models consider external factors such as weather [39] to improve accuracy. Zhang et al. [40] used deep multitask learning to forecast short-term energy demand of electricity, heat, and natural gas, considering the coupling between energy sources. Chen et al. [41] analyzed the energy consumption characteristics of a comprehensive energy system and constructed a load forecasting model based on stack ensemble learning, considering date factors. Zhao et al. [42] developed a load forecasting model for multiple loads using the bootstrap extreme learning machine method, considering historical loads, weather, dates, and social or economic indicators. The above studies mainly focus on improving algorithms to improve load forecasting accuracy. While these methods demonstrate predictive advantages, the predicted load remains volatile. Applying the predicted load directly as an input to load dispatch methods improves the foresight of load dispatch to some extent by understanding future demands in advance. However, the issue of equipment load volatility is not adequately addressed.

Therefore, it may be necessary to consider the possibility of flexible alignment of supply and demand matching on the timeline. Some researchers suggest using energy storage technologies [43] to mitigate the impact of demand fluctuations on production. Shaqsi et al. [44] discussed technologies such as flow batteries, supercapacitors, superconducting magnetic energy storage, flywheel energy storage, pumped storage, thermal energy storage, compressed air energy storage, and hybrid energy storage. Their work provides guidance for configuring energy storage in power grids and other systems. Amin Mirzaei et al. [45] investigated the role of thermal, electrical, and gas storage in the coordinated operation of power, natural gas, and district heating networks. Their results indicate that the introduction of energy storage can effectively reduce operating costs and mitigate the impact of multi-energy network constraints on system operation. In the hydro-thermal-wind-solar system proposed by Hemmati et al. [46], an energy storage system was incorporated to smooth out the fluctuations in renewable energy sources. Hussain et al. [47] integrated battery energy storage into the energy system to address the mismatch between photovoltaic power output and load demand during peak periods. Energy storages provide an intermediary role for flexible supply-demand matching, but require additional configuration costs.

Some researchers are contemplating whether there exist “flexible buffering equipment” within the current integrated energy systems that can achieve time buffering between supply and demand. As a result, some studies on “generalized energy storage” have begun to emerge. Boix et al. [48] defined the flexibility of a network as the minimum of the maximum deviations of the nominal input parameters that the network can handle, and used it for optimal network design. In the field of operational scheduling, Huang et al. [49] have highlighted the advantages of generalized energy storage resources and the thermal inertia of heat/cold loads in enhancing resilience. They argue that the considerable amount of hot water and gas stored in thermal and gas networks can be utilized over a certain period. Considering the difference in transmission speed between hot water and natural gas in pipelines and electricity, Chen et al. [50] have leveraged this energy time-shift characteristic to achieve pipeline energy storage. Furthermore, Hao et al. [51] have considered the role of flexible building loads in the flexible operation of smart grids. They have modeled it as a generalized battery to quantify flexibility. The aforementioned studies demonstrate that the inertia of fluid [52] and the presence of pipelines can provide a certain degree of flexibility for the operation of integrated energy systems, thereby reducing load fluctuations. This “generalized energy storage” is low-cost and naturally existing. However, there is currently limited research on the combination of generalized energy storage and load forecasting.

1.3. Motivation and contributions

Existing load dispatch research provides model support for the operation dispatch of integrated energy production systems in IP, offering optimized equipment load regulation strategies. However, there are some limitations in current research:

- 1) Untapped potential of system integration optimization: Existing studies have conducted research on load dispatch problems, achieving economic optimization, but neglecting how optimized load values are actually implemented. Load dispatch does not consider load regulation and prediction, losing some economic benefits and feasibility. Research on load regulation and load forecasting is relatively independent. Therefore, more attention needs to be paid to system-level integrated optimization to fully tap its potential.
- 2) Underutilization of load forecasting: Although existing studies have recognized the importance of advanced load forecasting for the regulation of production-side equipment, it often focuses too much on improving the accuracy of load forecasting and directly uses the predicted value for load dispatch input. Understanding load trends improves the feasibility of scheduling schemes to some extent, but it does not solve the problem of stability. In addition, although existing research considers stability measures such as configuring energy storage devices, few studies consider using generalized energy storage to buffer the supply and demand timetable, thereby optimizing the supply and demand matching mode.
- 3) Insufficient balance between economic benefits, stability, and executability: Existing studies have made progress in optimizing economic benefits, but lacks sufficient consideration for the stability and implementability of production. These studies often overlook the basic operation modes and habits required for equipment load regulation, and the fluctuation regulation strategies may be difficult to implement or may damage equipment life in the actual execution process. Therefore, how to balance economy and stability in load dispatch needs further research.

Table 1 provides an overview of the current research status on IP load smoothing and optimization.

This paper proposes a cross-level strategy of load smoothing regulation for IPs based on the data-driven method. The strategy utilizes the elasticity of the heat network as a buffer device, achieving flexible alignment of supply and demand based on load forecasting. At the same time, constraints for minimal changes in equipment load and

Table 1
Summary of research status of load smoothing and optimization in IPs.

Ref	Research modules			Method	Objective		Stability measures		
	Load forecasting	Load dispatch	Load regulation		Economy	Stationarity	Energy storage	Generalized energy storage	Improvement methods
[34]	✓	×	×	LS-SVM	×	×	×	×	×
[35]	✓	×	×	Temporal convolutional networks	×	×	×	×	×
[36]	✓	×	×	Encoder-decoder model based on LSTM	×	×	×	×	×
[41]	✓	×	×	Stacking ensemble learning	×	×	×	×	×
[39]	✓	×	✓	Deep learning and digital twins	✓	×	✓	×	Data augmentation method
[14]	×	✓	×	Distributed robust optimization	✓	×	✓	×	×
[16]	×	✓	×	PSO	✓	×	×	×	×
[17]	×	✓	×	GA	✓	×	×	×	×
[20]	×	✓	×	Energy stepped utilization	✓	×	✓	×	×
[23]	×	×	✓	GP	✓	×	×	×	×
[24]	×	×	✓	LS-SVM	✓	✓	×	×	✓
This study	✓	✓	✓	Cross-level optimization	✓	✓	×	✓	✓

operational parameters are introduced, achieving a balance between economy and stability. The main contributions of this paper are demonstrated as follows:

- 1) A cross-level strategy for load smoothing in IP that integrates load forecasting, dispatch, and execution is proposed, effectively mining the benefits of system integration optimization.
- 2) A method for utilizing load forecasting has been proposed. Building upon the foundation of a load forecasting model, the elasticity of the heat network is taken into account. This allows for the transformation of short-term demand predictions into energy supply predictions on the production side, thereby enhancing the stability of energy supply.
- 3) The importance of balancing economic efficiency and production stability has been considered. By introducing a minimum equipment load change constraint in the load allocation optimization model, a smooth distribution of supply to each equipment load is achieved. Further, by considering minimal changes in operational parameters during load regulation, a smooth implementation of equipment load is realized.

The remainder of this article is organized as follows. Section 2 describes the methodology followed in this work. This section introduces the structure of the IP studied in this work, along with the overall load smoothing method. Section 3 presents the case introduction and describes the cases setup. Section 4 introduces the obtained results and their discussion. The paper is concluded in Section 5 with a summary.

2. Methodology

The framework of the cross-level load smoothing and optimization method for IPs is depicted in Fig. 2. The model layer comprises three levels, namely load forecasting, load dispatch, and load regulation. The load forecasting level employs the LSTM model to predict the load, and the short-term forecasting model considers the date factor. The predicted load, buffered through heat network elasticity calculations, results in a stable forecast value for steam supply load. In the load dispatch level, a constraint for minimal equipment load changes is introduced to ensure a smooth distribution of equipment load. The load regulation level calculates the load regulation for individual equipment, aiming for minimal changes in operational parameters to achieve high-efficiency operation. The rolling load smooth regulation is carried out based on the actual steam supply and demand at the next moment.

2.1. Load forecasting model with smoothing

The predicted load time length is set as t_f min, and the time interval is defined as d_t min. The short-term load forecasting model is constructed

using LSTM model. The input parameter $\mathbf{X}$ is constituted by the temperature, pressure, quantity and time parameters in the past t_b time period.

$$\mathbf{X} = [\mathbf{p}, \mathbf{v}, \boldsymbol{\theta}, \mathbf{x}_m, \mathbf{x}_d, \mathbf{x}_h] \quad (1)$$

$$\mathbf{x}_m = \mathbf{x}_m / 12 \quad (2)$$

$$\mathbf{x}_d = \mathbf{x}_d / 31 \quad (3)$$

$$\mathbf{x}_h = \mathbf{x}_h / 24 \quad (4)$$

$$\mathbf{v}_p = f(\mathbf{X}) \quad (5)$$

Where, $\mathbf{p}, \mathbf{v}, \boldsymbol{\theta}, \mathbf{x}_m, \mathbf{x}_d, \mathbf{x}_h$ are pressure, quantity, temperature, months, days and hours in sequence; $\mathbf{X}$ is the input of LSTM model, the dimension of $\mathbf{X}$ is $(t_b, 6)$; $\mathbf{v}_p$ is the predicted load at t_f time points in the future, the dimension of $\mathbf{v}_p$ is $(1, t_f)$; f is a prediction model based on LSTM after training.

The predicted $\mathbf{v}_p$ presents a load with a relatively large fluctuation. Nevertheless, the steam network can be regarded as a vast storage tank that possesses certain flexibility and adjustment capacity. The pipe network pressure is subject to increase when the inlet steam flow exceeds the outlet steam flow, while it falls when the inlet steam flow falls short of the outlet steam flow. Upon hitting the preset pressure boundary, the traditional controller control logic automatically triggers control actions, such as adapting the valve opening of the steam inlet and outlet to maintain the pipeline pressure within a stable range. Hence, the difference between the current steam supply flow and the predicted user demand can be analyzed using the steam supply capacity of the pipe network, and the equipment load can be kept to the greatest extent possible while ensuring the pressure safety of the pipe network.

$$\mathbf{p}_s = \mathbf{p}_r - (\mathbf{v}_p - \mathbf{v}_s) \times d_t \times \boldsymbol{\theta}_r \times R / (M \times V_{\text{pipe}}) \quad (6)$$

$$\mathbf{L}_{vs} \leq \mathbf{v}_s \leq \mathbf{U}_{vs} \quad (7)$$

$$\mathbf{v}_{s,0} = \mathbf{v}_r \quad (8)$$

Where, $\mathbf{p}_s$ is the pressure of pipe after smooth, which needs to be maintained within the safe range; $\mathbf{p}_r$ is the actual pressure of the pipe at the time of prediction; $\mathbf{v}_s$ is the quantity of pipe after smooth, which is the result we want to get; $\boldsymbol{\theta}_r$ is the actual temperature of the pipe; V_{pipe} is the capacity of the pipe; $\mathbf{v}_{s,0}$ is the initial value of $\mathbf{v}_s$; $\mathbf{v}_r$ is the actual quantity of the pipe; $\mathbf{L}_{vs}$ is the lower limit of pipe quantity; $\mathbf{U}_{vs}$ is the upper limit of pipe quantity; M is molar mass; R is the gas constant.

The stability of the pipeline network pressure is guaranteed by implementing a slight range of upward and downward adjustments. The maximum upper and lower limits of such regulation are constrained by the load regulation capacity of the production equipment, for example, boilers.

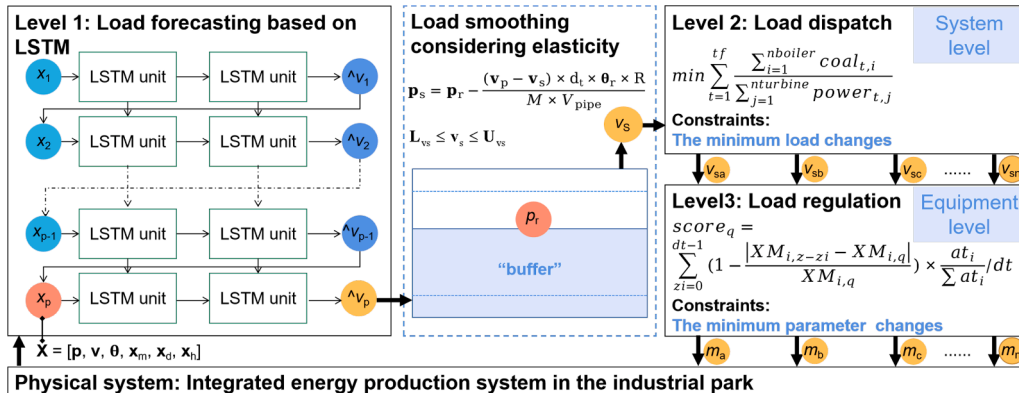


Fig. 2. The framework of the cross-level load smoothing and optimization method.

2.2. Load dispatch model with smoothing

Following load forecasting and smoothing calculation, the obtained total steam supply is relatively stable. Nevertheless, multiple boilers and steam turbines are typically equipped in integrated energy production systems. Different operating modes of boilers and steam turbines can generate identical total products, yet their economy, environmental protection, and energy efficiency can differ. To address these challenges, this paper proposes a load dispatch optimization model based on load smoothing. The proposed model aims to minimize the standard coal consumption rate for generation, while considering optimized load stability for each boiler. Furthermore, the model optimizes and calculates the loads of each equipment in the integrated energy production system, achieving smooth load distribution at the equipment level. The objective function and constraints of the load dispatch model with smoothing are as follows:

Objective function:

$$\min \sum_{t=1}^{T_t} \sum_{i=1}^{n_{\text{boiler}}} v_{\text{coal},t,i} / \sum_{j=1}^{n_{\text{turbine}}} P_{t,j} \quad (9)$$

Where, n_{boiler} is the total number of boilers; n_{turbine} is the total number of turbines; $v_{\text{coal},t,i}$ is the coal consumption of boiler i at time t ; $P_{t,j}$ is the generating capacity of steam turbine j at time t .

Constraints:

1) Boiler related constraints

$$v_{\text{coal},t,i} = v_{\text{bs},t,i} \times H_{\text{b},t,i} / (H_{\text{coal},t,i} \times E_{\text{b},t,i}) \quad (10)$$

$$L_{\text{coal},i} \leq v_{\text{coal},t,i} \leq U_{\text{coal},i} \quad (11)$$

$$L_{\text{bs},i} \leq v_{\text{bs},t,i} \leq U_{\text{bs},i} \quad (12)$$

$$L_{\text{dbs},i} \leq |v_{\text{bs},t+1,i} - v_{\text{bs},t,i}| \leq U_{\text{dbs},i} \quad (13)$$

Where, $v_{\text{bs},t,i}$ is the steam production of boiler i at time t ; $H_{\text{b},t,i}$ is the enthalpy value of boiler steam i at time t ; $H_{\text{coal},t,i}$ is the enthalpy value of feed coal of boiler i at time t ; $E_{\text{b},t,i}$ is the efficiency of boiler i at time t ; $L_{\text{coal},i}$ is the lower limit of coal consumption of boiler i ; $U_{\text{coal},i}$ is the upper limit of coal consumption of boiler i ; $L_{\text{bs},i}$ is the lower limit of steam production of boiler i ; $U_{\text{bs},i}$ is the upper limit of steam production of boiler i ; $L_{\text{dbs},i}$ is the lower limit of ramping; $U_{\text{dbs},i}$ is the upper limit of ramping

2) Turbine related constraints

$$L_{\text{tin},j} \leq v_{\text{tin},t,j} \leq U_{\text{tin},j} \quad (14)$$

$$L_{\text{tout0},j} \leq v_{\text{tout0},t,j} \leq U_{\text{tout0},j} \quad (15)$$

$$L_{\text{tout1},j} \leq v_{\text{tout1},t,j} \leq U_{\text{tout1},j} \quad (16)$$

$$P_{t,j} = a_j \times v_{\text{tin},t,j} + b_j \times v_{\text{tout0},t,j} + c_j \times v_{\text{tout1},t,j} + d_j \quad (17)$$

$$L_{\text{p},j} \leq P_{t,j} \leq U_{\text{p},j} \quad (18)$$

Where, $v_{\text{tin},t,j}$ is the steam inlet of turbine j at time t ; $L_{\text{tin},j}$ is the lower limit of the steam inlet of turbine j ; $U_{\text{tin},j}$ is the upper limit of the steam inlet of turbine j ; $v_{\text{tout0},t,j}$ is the extraction flow of turbine j at time t ; $L_{\text{tout0},j}$ is the lower limit of extraction flow of turbine j ; $U_{\text{tout0},j}$ is the upper limit of extraction flow of turbine j ; $v_{\text{tout1},t,j}$ is the exhaust flow of turbine j at time t ; $L_{\text{tout1},j}$ is the lower limit of exhaust flow of turbine j ; $U_{\text{tout1},j}$ is the upper limit of exhaust flow of turbine j ; a_j, b_j, c_j, d_j are the power generation model coefficients of turbine j ; $L_{\text{p},j}$ is the lower limit of power generation of turbine j ; $U_{\text{p},j}$ is the upper limit of power generation of turbine j .

3) Balance constraints

$$\sum_{i=1}^{n_{\text{boiler}}} v_{\text{bs},t,i} = \sum_{j=1}^{n_{\text{turbine}}} v_{\text{tin},t,j} + \sum_{z=1}^{n_{\text{compressor}}} v_{\text{cin},t,z} + \sum_{q=1}^{n_{\text{tdpr0}}} v_{\text{aout},t,q} + \sum_{q=1}^{n_{\text{tdpr1}}} v_{\text{aout},t,q} \quad (19)$$

$$\sum_{j=1}^{n_{\text{turbine}}} v_{\text{tout0},t,j} + \sum_{q=1}^{n_{\text{tdpr0}}} v_{\text{aout},t,q} = v_{\text{sup0},s,t} \quad (20)$$

$$\sum_{j=1}^{n_{\text{turbine}}} v_{\text{tout1},t,j} + \sum_{q=1}^{n_{\text{tdpr1}}} v_{\text{aout},t,q} = v_{\text{sup1},s,t} \quad (21)$$

Where, $n_{\text{compressor}}$ is the total number of compressors; $v_{\text{cin},t,z}$ is the steam inlet of compressor z at time t ; $v_{\text{sup0},s,t}$, $v_{\text{sup1},s,t}$ are the total amount of different steam supply after load forecast smoothing at time t , which are subsets of v_s ; n_{tdpr0} is the total number of low-pressure TDPRs; n_{tdpr1} is the total number of medium-pressure TDPRs; $v_{\text{aout},t,q}$ is the steam output of the TDPR q at time t .

4) Smoothing constraints

To minimize load fluctuation, the number of boilers that require load adjustments is recorded as n_{change} . If the boiler load change between the last and the proposed time is greater than 0, it indicates that the boiler load has been adjusted during that time.

Algorithm 1 is the pseudo-code for the load dispatch optimization model that takes into account equipment load smoothing.

2.3. Load regulation model based on pattern-matching

The load dispatch optimization calculation results only apply to the equipment load level, and it does not provide specific operational methods to achieve the recommended optimal load. Different operation modes correspond to different equipment efficiency, hence there is a need to calculate an effective and specific load regulation method based on the current equipment operational mode. This paper adopts the boiler optimization model based on pattern matching. The construction method of this specific model has been published in another article [53]. Algorithm 2 is the pseudo-code.

3. Case introduction

This study takes an IP in Zhejiang Province, China, as an example. The integrated energy production system in the IP is equipped with four sets of 220 t/h high temperature and ultra-high pressure circulating fluidized bed boilers, two sets of 30 MW back-type steam turbine generator units, and two sets of 1500 Nm³/min back-pressure steam turbine compressed air units. The IP primarily serves printing and dyeing and textile enterprises, which are the main steam users. Specifically, the IP serves 33 medium-pressure steam users and 211 low-pressure steam users, with an annual heating capacity of 19.63 million tons and 3.18 billion kWh.

To train the short-term load forecasting model, the total amount of medium and low-pressure steam data from August 12, 2021, 17:06:41 to January 7, 2022, 23:54:06 is selected. Fig. 3 depicts the steam demand

Algorithm 1

Load dispatch optimization considering equipment load smoothing.

Input: smoothed steam load supply.

Output: optimized load of boiler, steam turbine and other equipment.

1) Construct the optimization model of inclusion Eq. 9-21.

2) $n_{\text{change}} = 1$.

3) Call the optimization model in Step 1 to solve. If there is a feasible solution, skip to Step 5), otherwise skip to Step 4).

4) $n_{\text{change}} = n_{\text{change}} + 1$, and repeat step 3).

5) Result output.

Algorithm 2

Load regulation model for boilers based on pattern matching.

Input: current coal quality, main steam temperature, main steam pressure, feed-water temperature, target steam flow, boiler efficiency, frequency conversion of coal feeder and fan operation.
Output: specific oxygen and air volume data of coal feeder and fan operation frequency conversion and control.

- 1) Integrate the current coal quality, main steam temperature, main steam pressure, feed-water temperature, and target steam flow data, and label them as working condition parameters A_1 .
- 2) Specify the minimum similarity tolerance lb .
- 3) Employ a distance measure to identify the historical time points of the boiler database which possess similarity measures that exceed lb , while operating at a greater efficiency than the current time. Define the resulting time points as set R_1 .
- 4) Upon formulating set R_1 , progressively screen the higher efficiency boiler segments by following the bed temperature, oxygen concentration, primary and secondary air ratio, primary air volume, and secondary air volume sequence to obtain the set R_2 .
- 5) Utilizing R_2 , compute the similarity scores between the frequency conversion data of the coal feeder and fan operation. Determine the time point scheme with the highest similarity measure as the final result R_3 .

trends for medium and low-pressure steam (with the first 60000 pieces of data).

The upper and lower limits of pipe network pressure are set as shown in Table 2.

Table 3 displays the upper and lower limits of the load. It's important to note that the lower limit indicated in Table 3 denotes the minimum load capacity of the equipment when it's powered on.

4. Results and discussions

Utilizing the aforementioned case as a basis, this study initially evaluates the effectiveness of each level model. Subsequently, the superiority of the proposed methodology in comparison to manual operations is affirmed. Lastly, a comparison between the proposed methodology and traditional approaches is conducted, accompanied by a thorough analysis of the significance of each level of smoothing through contribution and necessity analysis. At the same time, the response of the proposed method to the uncertainty of load forecasting is discussed.

4.1. Validation of basic models at three levels

4.1.1. Level 1: load forecasting model

The training and test data were divided into a ratio of 7:3. Two separate models were developed to predict the demand of medium-pressure and low-pressure steam demands for the next half hour. The assessment of the prediction models' accuracy was conducted on the test set utilizing the Mean Absolute Percentage Error (MAPE) metric. The

Table 2

Network pressure configuration parameters.

Steam pipe network	Maximum pressure (MPa)	Minimum pressure (MPa)	The highest in history (MPa)	The lowest in history (MPa)	Volume (Nm ³)
Medium-pressure steam	3.20	3.05	3.41	2.92	2999.43
Low-pressure steam	0.90	0.70	0.97	0.65	16598.48

Table 3

Load constraint settings.

Item	Lower limit	Upper limit	Unit
Load of boiler #1	90	220	t/h
Load of boiler #2	90	220	t/h
Load of boiler #3	90	220	t/h
Load of boiler #4	90	300	t/h
Inlet steam volume of turbine #1	90	304.5	t/h
Inlet steam volume of turbine #2	90	304.5	t/h
Power generation of turbine #1	5	35	MW
Power generation of turbine #2	5	35	MW
Steam exhaust of turbine #1	50	240	t/h
Steam exhaust of turbine #2	50	240	t/h
Steam extraction of turbine #1	0	80	t/h
Steam extraction of turbine #2	0	80	t/h
TDPR #1	0	960	t/h
TDPR #2	0	960	t/h
TDPR #3	0	960	t/h
TDPR #4	0	960	t/h
TDPR #5	0	960	t/h

obtained MAPE values for medium-pressure and low-pressure user demand were 2.44% and 1.68%, respectively, satisfying the requirements for industrial load forecasting. Further, a comparison was made between the direct time series-based prediction, the hybrid prediction method without considering the date factor, and the hybrid time series and feature prediction method proposed in this study that considers steam temperature, pressure, and date factors.

Table 4 shows the results of various indicators of load forecasting. Specifically, whether it is the prediction of low-pressure steam load or medium-pressure steam load, the prediction error indicators obtained by directly using the time series-based method are the worst. After adding the information of steam temperature and pressure over a period of time, the prediction effect has been improved, and the mean absolute Error (MAE) of medium and low-pressure prediction has improved by 9.80% and 12.78% respectively. Furthermore, after adding the date factor,

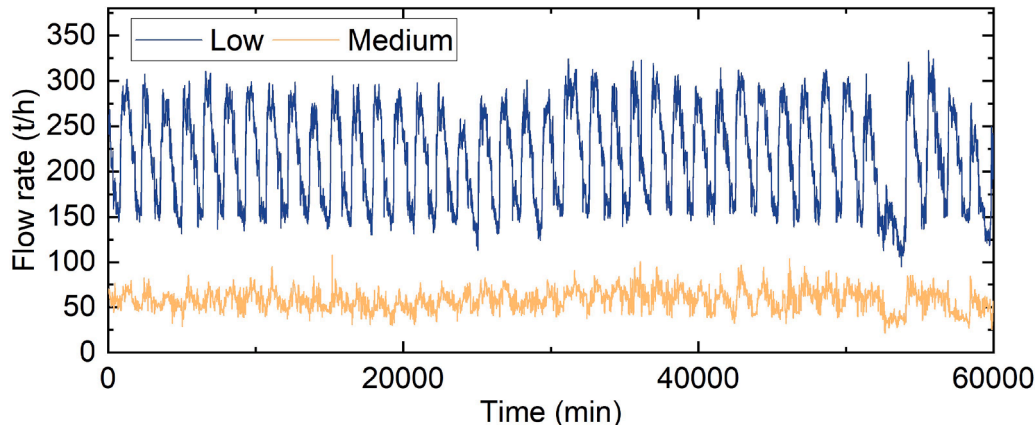


Fig. 3. Historical trend of medium and low-pressure steam demand.

Table 4
Indicator results of medium and low-pressure steam load forecasting.

Method	Medium-pressure					Low-pressure				
	MAPE (%)	MAE (t/h)	MSE (t^2/h^2)	RMSE (t/h)	R^2	MAPE (%)	MAE (t/h)	MSE (t^2/h^2)	RMSE (t/h)	R^2
Time-series-based method	2.55	3.63	30.29	5.50	0.83	2.20	4.38	43.46	6.58	0.95
Hybrid method (without data information)	2.30	3.28	23.62	4.86	0.87	1.92	3.82	31.38	5.59	0.96
Proposed hybrid method	2.44	2.56	13.33	3.65	0.90	1.68	3.14	19.41	4.40	0.98

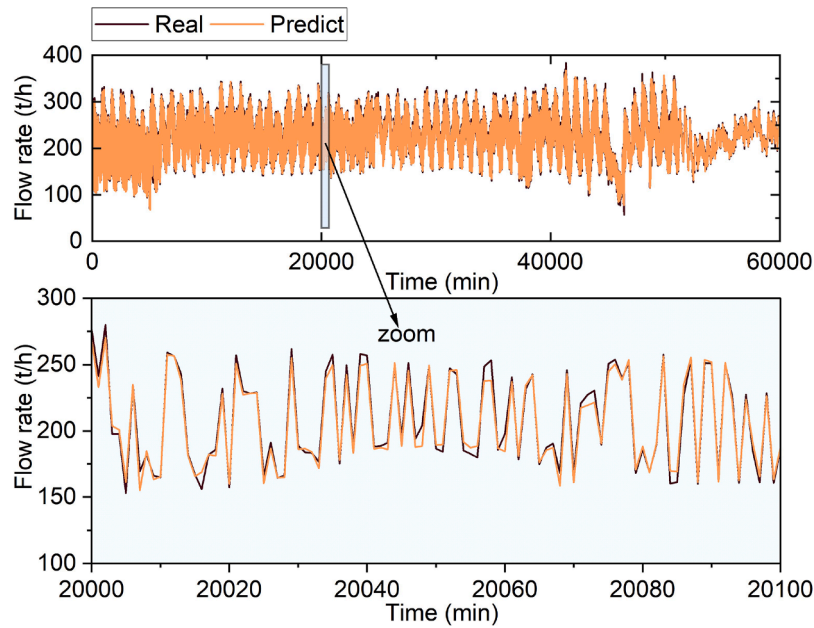


Fig. 4. Comparison between actual and predicted low pressure steam demand.

although the MAPE of medium pressure has increased to some extent, the MAE, mean square error (MSE), root mean square error (RMSE), and R^2 have all improved. The MAE of medium and low-pressure prediction has improved by 21.95% and 17.80% respectively compared to not considering the date factor.

Through these comparative experiments, the method of this study, based on the hybrid method, effectively improves the accuracy of load forecasting by integrating temperature, pressure, and date features.

Fig. 4 and Fig. 5 illustrate the comparison between the predicted and actual values of the low-pressure and medium-pressure user demand

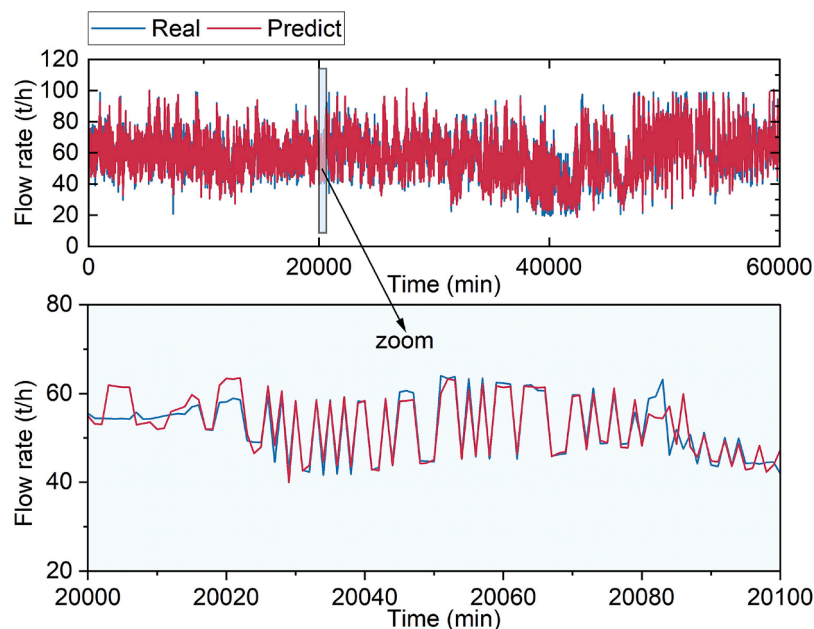


Fig. 5. Comparison between actual and predicted medium pressure steam demand.

Table 5
Three typical operating conditions.

Case	Date	Equipment status					
		Boiler #1	Boiler #2	Boiler #3	Boiler #4	Turbine #1	Turbine #2
Case #1	2022/05/07	✓	×	✓	✓	✓	×
Case #2	2022/06/01	×	✓	×	✓	✓	×
Case #3	2022/10/19	✓	✓	✓	×	✓	×

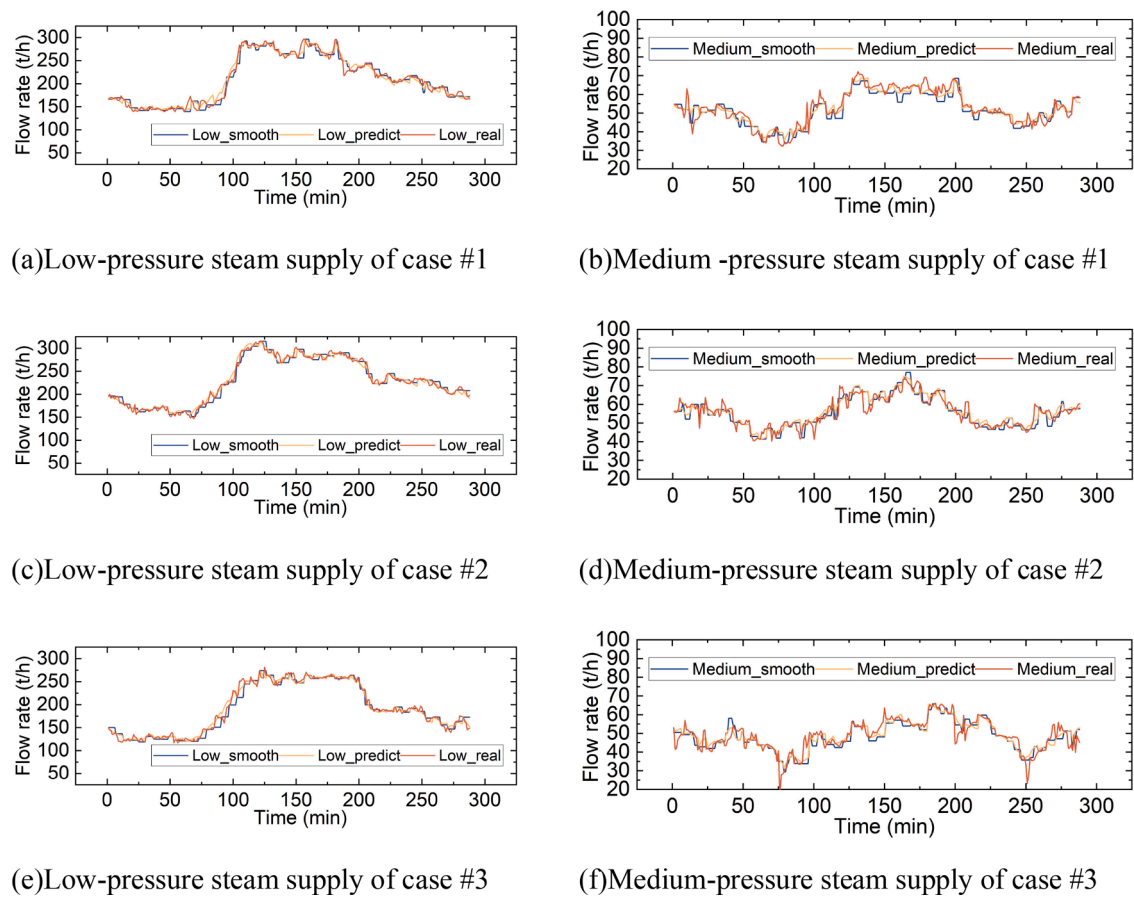


Fig. 6. Smooth steam supply on typical days.

(with the first 60000 pieces of data). The results indicate that low-pressure steam demand surpasses medium-pressure steam demand and exhibits fluctuations. Except for some individual spikes, the accuracy of the predicted values shows good agreement with the actual values.

To further verify the effectiveness of smooth steam supply under diverse operating conditions, three standard daily work cases in [Table 5](#)

are examined and analyzed, and the results are depicted in [Fig. 6](#).

The outcomes demonstrate that the low-pressure steam supply curves exhibit a distinguishable pattern that remains high at the medium range and low at both ends across all operating conditions. Conversely, the medium-pressure steam supply curves display multiple peaks. Notably, the predicted curves align closely with the actual curves, albeit

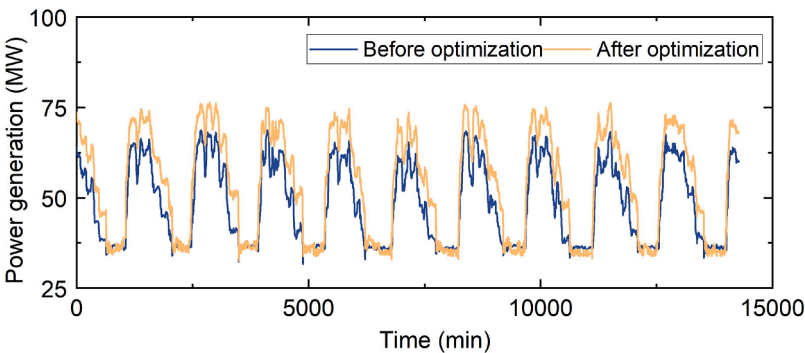


Fig. 7. Comparison of power generation before and after load optimization.

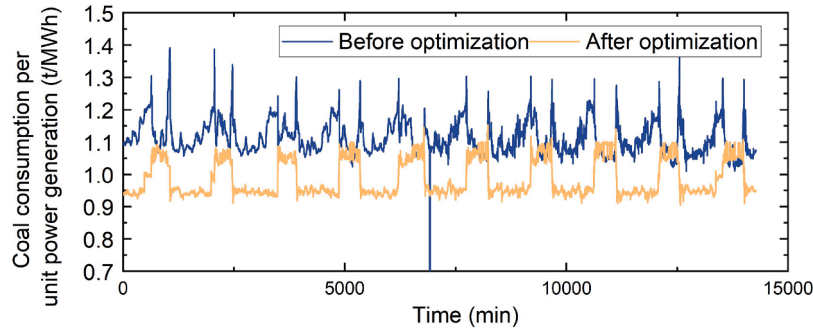


Fig. 8. Comparison of coal consumption per unit power generation before and after load optimization.

featuring some small ladder shapes in the otherwise smooth curves. In addition, the smoothed curves effectively mitigate abrupt flow fluctuations, as manifested by their consistent trajectory relative to the measured curves. The above results indicate that the load forecasting model adopted in this article is effective.

4.1.2. Level 2: load dispatch model

The load dispatch optimization computation outcomes from May 18 to May 28, 2021, are illustrated in Fig. 7 and Fig. 8. The generating capacity of the unit exhibits a certain periodicity, with the lowest capacity recorded at approximately three o'clock in the morning. Thereafter, the unit's capacity gradually rises until it reaches the maximum value at approximately 1 o'clock in the afternoon, and subsequently undergoes a gradual reduction.

Upon the implementing the load dispatch model proposed in this study, significant improvements in the unit's power generation performance have been observed, with an average daily improvement value of 2.35 MW. This is because the load dispatch method proposed in this study aims to minimize the coal-to-electricity ratio in the IP. Under the constraints of boilers, turbines, energy balance, and stable scheduling, it can achieve the optimal equipment load with the minimum coal-to-electricity ratio. In contrast, manual decision-making relies on experience, and it is difficult to make decisions when coordinating optimization of multiple units within the IP. Operations based on manual operations are feasible, but not optimal. Additionally, the optimized coal consumption rate per power generation unit also reports a substantial decrease, with an average daily reduction of 0.097 t/MWh, as depicted in Fig. 8. These results demonstrate the effectiveness of our proposed method in enhancing the operational efficiency of the power generation unit.

4.1.3. Level 3: equipment load regulation model

The optimization method proposed in this study exhibits a significant enhancement in efficiency for the boiler operations under different loads, with minimal changes in the operating parameters. At a load of 191 t/h, the method enhances boiler efficiency by 1.92%, whereas under the fundamental operational settings of 178 t/h, the efficiency can be increased by 2.14%.

To exemplify, an assessment of the 178 t/h operating condition is performed, and the outcomes of parameter optimization for boiler bed temperature, oxygen concentration, as well as primary and secondary air ratio, are presented in the Figs. 9–11. The figures demonstrate that, at a load of 178 t/h, the optimal bed temperature range is between 827.17 and 877.80 °C, whereas the recommended oxygen concentration ranges from 1.90 to 2.65%. Additionally, the optimal ratio of primary and secondary air falls within the range of 1.16 to 1.82.

To sum up, the proposed hierarchical models in this study, including the load forecasting, load dispatch, and equipment load regulation models, have all been verified to be effective.

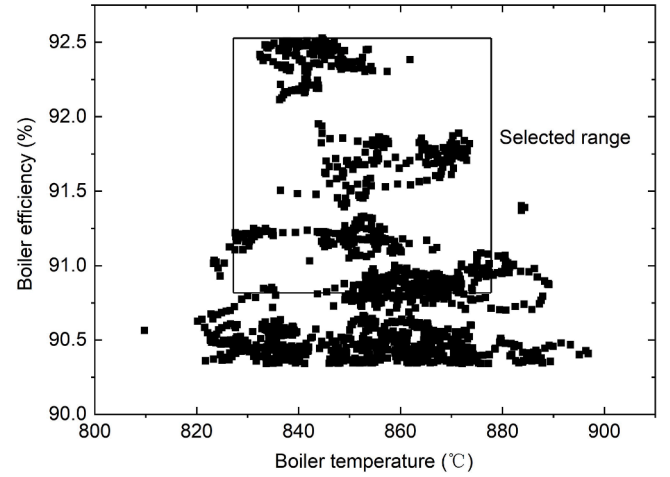


Fig. 9. Optimization process under typical conditions-boiler temperature.

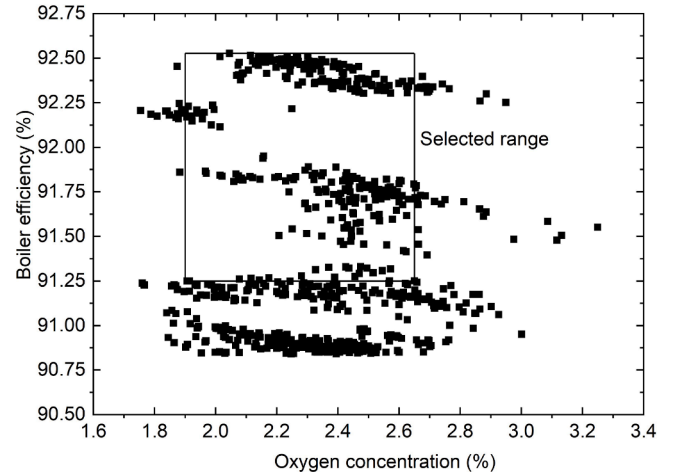


Fig. 10. Optimization process under typical conditions-boiler oxygen concentration.

4.2. Overall load optimization results under different working conditions

To illustrate the superiority of the proposed method over manual operations, various time points throughout the day are selected, and the optimization approach suggested in this study is employed for calculations. Table 6 illustrates that during periods of low user demand, steam supply decreases from 00:00–6:00 and rapidly rises in demand to peak at midnight when user load is high. There is a slight drop in steam supply at 18:00, but it remains at a relatively high load level. Using the proposed

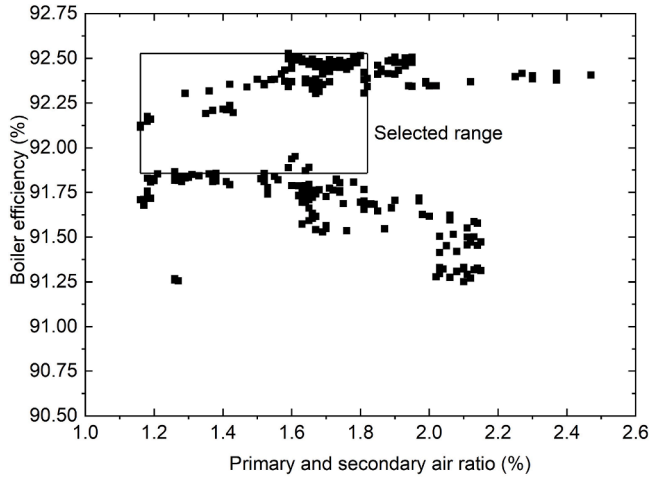


Fig. 11. Optimization process under typical conditions-primary and secondary air ratio.

optimization method, the coal-electricity ratio can be decreased for different operating conditions, with an average reduction of 21.46%, alongside an average increase in power generation of 23.45% across the four time periods analyzed in this case. Additionally, the optimization method has enhanced boiler efficiency, exhibiting an overall average increase of 0.59%.

The above results support the effectiveness of the proposed industrial cross-level load smoothing optimization method. The decrease in the coal-electricity ratio can be explained by the optimized dispatch of equipment, which decreased coal consumption and increased power generation, leading to more efficient use of resources.

4.3. Discussions

4.3.1. Comparison between our smoothing method and traditional methods

To provide a more comprehensive demonstration of the effectiveness of the proposed method on load management, a smoothness index calculation is introduced as an objective measure of the stability and consistency of the load. The non-smoothness index (NSI) calculation is defined as the average of the absolute differences between adjacent data points divided by the mean value of the data, which effectively evaluates load smoothness.

Specifically, denote the load data as D ($D = [a_1, a_2, a_3, \dots, a_n]$), with its mean value represented as D_m . The NSI of the load data can be quantified using the following formula:

$$NSI = \frac{\sum_{i=1}^n |a_i - a_{i-1}|}{n \times D_m} \times 100 \quad (22)$$

Table 6

Optimization results of the method proposed under different operating conditions.

Item	2023/4/1 0:00		2023/4/1 6:00		2023/4/1 12:00		2023/4/1 18:00	
	Actual	Optimal	Actual	Optimal	Actual	Optimal	Actual	Optimal
Medium-pressure steam supply (t/h)	37.36	34.01	34.62	32.71	61.02	62.04	51.46	54.88
Low-pressure steam supply (t/h)	182.03	185.05	156.28	159.95	222.19	230.56	218.05	225.85
Main steam #1 (t/h)	111.59	111.60	97.25	103.90	132.94	136.08	124.27	124.13
Main steam #1 (t/h)	108.98	107.08	95.75	103.90	122.82	123.03	114.26	114.77
Main steam #3 (t/h)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Main steam #4 (t/h)	135.55	135.82	114.89	124.04	202.45	202.10	197.66	206.14
Efficiency of boiler #1 (%)	89.93	90.52	89.92	90.41	90.71	90.84	89.91	90.67
Efficiency of boiler #2 (%)	88.73	88.98	88.47	88.74	88.80	90.40	88.64	90.19
Efficiency of boiler #3 (%)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Efficiency of boiler #4 (%)	91.54	91.81	91.02	90.97	91.97	92.15	91.74	92.04
Total coal consumption (t/h)	48.14	44.92	42.18	43.93	61.19	53.69	57.40	51.74
Total power generation (MW)	24.53	25.39	14.66	25.39	36.99	40.34	36.73	39.69
Ratio of coal to electricity (t/MWh)	1.96	1.77	2.88	1.73	1.65	1.33	1.56	1.30

The smoothness index (SI) is defined as:

$$SI = 100 - NSI \quad (23)$$

To evaluate the impact of the proposed method on load smoothness, three typical cases in Table 5 were calculated. The results of steam supply stability before and after smoothing are shown in Table 7.

The analysis results show that the proposed method significantly improved load smoothness, with the average smoothness of medium-pressure and low-pressure pipes improving by 33.95% and 36.48%, respectively.

Utilizing data collected from May 18th to May 28th, 2021, a load dispatch optimization method that does not consider load smoothing is developed. As presented in Table 8, the results demonstrate that regardless of whether load smoothing is considered, the optimization method outperforms manual operation regarding average power generation and coal consumption. Specifically, the method results in an average reduction of 0.92 t/h in coal consumption, an average increase of 5.69 MW in power generation, and an average decrease of 10.81% in the coal-to-electricity ratio when compared to manual operation.

Comparing the traditional optimization method without load

Table 7

Smoothness of medium and low-pressure steam supply.

Low-pressure steam	NSI		Medium-pressure steam	NSI	
	Before	After		Before	After
Case #1	1.62	1.35	Case #1	2.77	2.10
Case #2	1.79	0.94	Case #2	3.41	2.30
Case #3	1.59	0.87	Case #3	2.86	1.57

Table 8

Comparison of methods with and without load smoothing.

Item	Manual operation	Our proposed optimization method with smoothing	Traditional non-smooth optimization
Boiler load variation	3	1	3
Coal consumption per ton of steam (t/t)	0.13	0.12	0.12
Coal consumption (t/h)	53.22	52.30	52.08
Power generation (MW)	48.09	53.78	53.49
Average ratio of coal to electricity (t/MWh)	1.11	0.99	0.99
Boiler steam production (t/h)	436.30	428.76	427.43

Table 9

Necessity verification of each level model - method design.

Method name	Equipment regulation	Load forecasting	Steam supply smoothing	Load dispatch
Method 1 (Basic method)	✓	✓	✓	✓
Method 2	×	✓	✓	✓
Method 3	✓	×	×	✓
Method 4	✓	✓	×	✓
Method 5	✓	✓	✓	×

smoothing to our developed method, the latter allows for the control of load variation to one boiler which significantly reduces the operation volume. Additionally, after conducting a condition optimization calculation, the boiler operation similarity exceeds 95%. These results demonstrate the superiority of our unit optimization method and its potential to improve power generation efficiency.

4.3.2. Analysis of the contribution and necessity of each level model

To investigate the proposed method's effectiveness and necessity, the impact of each level model on the overall scheme is analyzed. As illustrated in Table 9, five verification methods are designed. Method 1, which serves as the baseline approach, employs the proposed cross-level optimization method. Method 2 omits load regulation to test its impact on the overall dispatch scheme. Method 3 disregards load forecasting and smoothing. Method 4 solely excludes load smoothing, and Method 5 does not take into account load distribution.

Fig. 12 illustrates the impact of each method on the average coal-to-electricity ratio, power generation, boiler efficiency, and main steam supply.

The results of the analysis lead to the following conclusions:

- 1) In terms of average coal-to-electricity ratio improvement, Method 3 exhibits the highest improvement, followed by Method 4 and Method 1. Nevertheless, Method 3's largest boiler load change indicates a trade-off with smoothness.
- 2) In terms of average power generation improvement, Method 1 is superior to other methods in optimizing the system's power generation.
- 3) Each level model has the ability to improve boiler efficiency when compared to manual operation. Nonetheless, Method 1's actual

boiler efficiency is lower than that of Method 2 when operational similarity is considered. This demonstrates the challenge of fully utilizing load optimization effectiveness through traditional load dispatch approaches without execution optimization. Furthermore, Method 4 exhibits slightly higher average boiler efficiency improvement than Method 1 because of significant total load change in the boiler without load smoothing.

- 4) Method 1 effectively improves smoothness regarding the average change in the main steam supply.

The research outcomes demonstrate the importance and necessity of integrating and smoothing various levels, including load forecasting, load dispatch, and equipment regulation, for optimizing the energy utilization efficiency of complex systems. Despite the pros and cons of various methods, the recommended inter-level load streamlining technique reveals encouraging outcomes in balancing load optimization effectiveness and system smoothness. From the contribution perspective, the load dispatch level has the most significant improvement effect on the overall system's energy efficiency and economic performance. However, it should be noted that implementing load dispatch results depends on the downstream equipment regulation module. The premise of load dispatch is limited by upstream demand forecasting.

4.3.3. Analysis of uncertainty

Furthermore, in order to analyze the advantages of the method proposed in this study in dealing with uncertainty, the impact of uncertainty disturbances has been examined. Specifically, the effect of demand forecasting fluctuations on steam load has been considered. A random term has been added to the load forecast value and the results have been tested under three different cases.

$$d' = \bar{d} \times (r + 1) \quad (24)$$

Where, d' is the predicted load value after interference; $\bar{d}$ is the average of the original half hour load forecasting value; r is a random value in the range of 0-0.1, following a uniform distribution.

Fig. 13 illustrates that random interference directly alters the predicted value. In the absence of the heat network's flexibility serving as a buffer device, the disturbed predicted value is directly utilized as the total steam demand and conveyed to the load dispatch module. Nevertheless, with the implementation of the method proposed in this study which considered of the heat network elasticity, the smoothed load

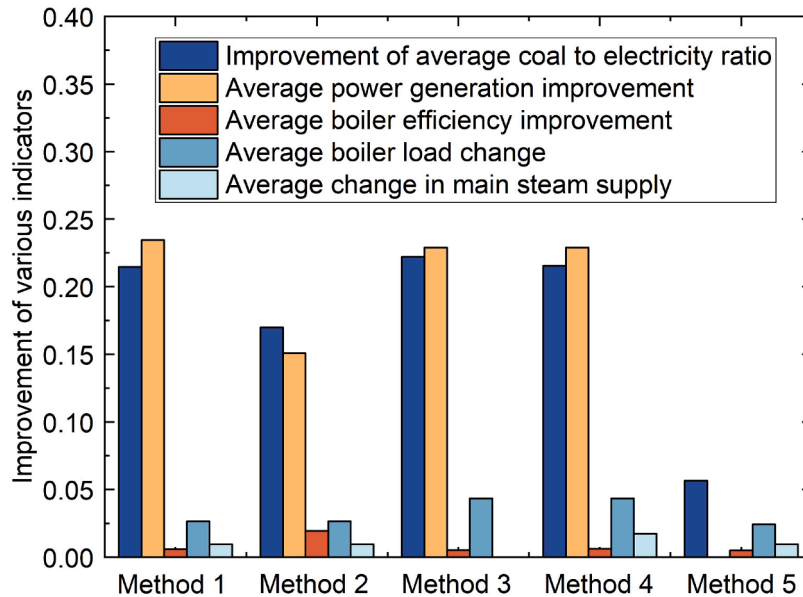


Fig. 12. Improvement of scheme indicators in different methods.

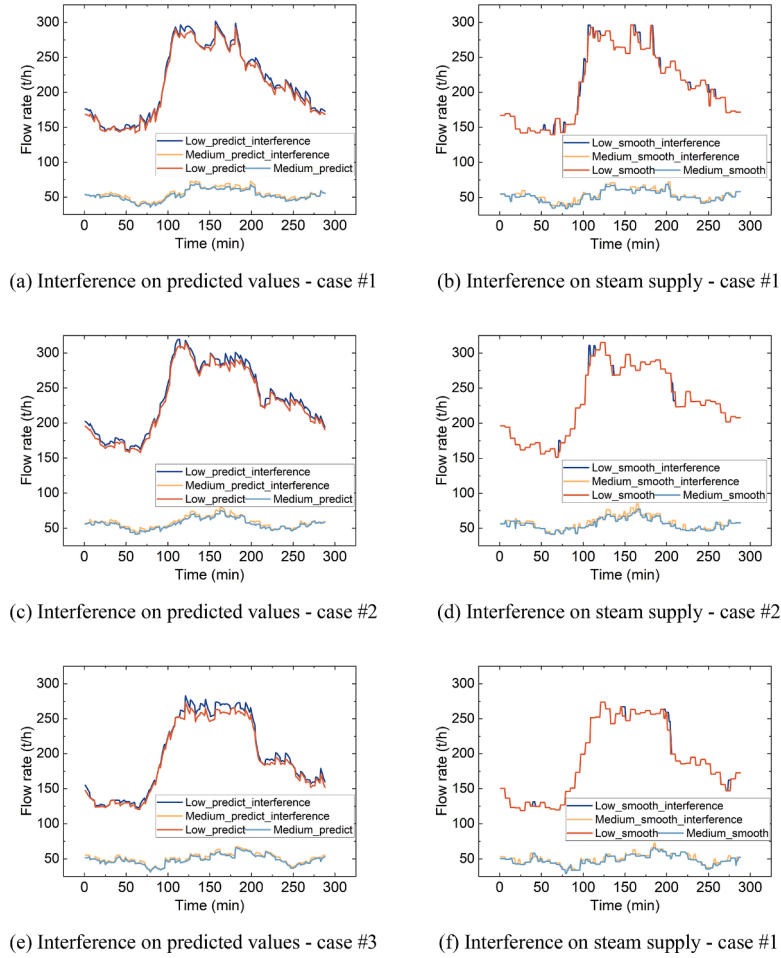


Fig. 13. The impact of demand forecasting interference on steam supply load.

value can still approximate the value prior to the disturbance to a certain extent, even in the presence of interference in the predicted value.

A consistency parameter is introduced herein to quantify the impact of load forecast disturbances on steam demand. The consistency parameter, denoted as CI, signifies the average rate of unchanged steam forecast values before and after the disturbance in demand load forecasting over a specific period.

$$CI = \frac{\sum_{i=0}^n I(v_{1,i}, v_{0,i})}{n} \times 100 \quad (25)$$

Where, CI is the consistency parameter; n is the total time length; v_1 is the steam supply after interference; v_0 is the steam supply before interference; I is the judgment function, if $v_{1,i} = v_{0,i}$, it is 1, otherwise it is 0.

As shown in the Table 10, the coincidence parameter of low-pressure steam is higher, and the test results of the three cases can reach a minimum of 87%, while the coincidence parameter of medium-pressure steam is about 50%.

Additionally, the impact of random interference at four typical moments on load dispatch has been tested, as depicted in Fig. 14. The findings indicate that following a fixed increase of 10% in the total steam supply increment disturbance, the total boiler load increase, without considering equipment load changes, exceeds the minimum load changes considered by the method proposed in this study, with an average increment of 11.64 t/h. Concurrently, when the increase in random disturbance ranges between 0-10%, upon implementing the method proposed in this study, the change in total steam supply instigated by random disturbance only influences the load of individual equipment, whereas the traditional method that does not take into

Table 10

The impact of demand forecasting interference on the consistency of steam supply.

Case	CI (%)	
	Medium-pressure	Low-pressure
Case #1	52.08	87.15
Case #2	94.10	54.51
Case #3	95.49	48.61

account the minimum equipment load change will interfere with the load of all equipment. This further demonstrates that considering the stable operation of equipment can mitigate the impact of source-load uncertainty to a certain extent.

In conclusion, the study highlights the significance and urgency of embracing a comprehensive cross-level strategy to maximize the energy utilization efficiency of intricate systems. The proposed approach exhibits immense potential for application in energy management systems by streamlining the synergy between diverse levels, balancing economic and stability, and attaining optimal performance.

5. Conclusion

It is essential to fully tap the knowledge behind the data, improve the energy efficiency utilization of the IP, and stabilize production. This paper proposes a cross-level steam load smoothing and optimization method for IPs. The proposed method concentrates on the equipment level and utilizes the elasticity of the heat network to regulate steam supply load while meeting user demands effectively, achieving flexible

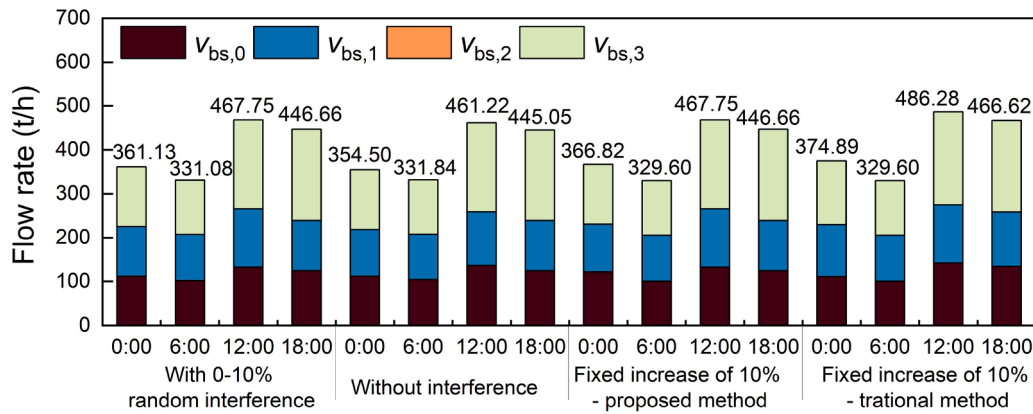


Fig. 14. The impact of random interference at four typical moments on load dispatch.

alignment of supply and demand. The results of experiments demonstrate that the proposed method significantly smooths medium and low-pressure loads and increases the average power generation by 5.69 MW. Moreover, the pattern-matching optimization technique enhances efficiency by increasing boiler efficiency by 2.14% at a typical load of 178 t/h. Furthermore, the findings authenticate the significance and effectiveness of integrating load smoothing across all levels and emphasize the importance of load dispatch for optimizing system efficiency as a whole.

Overall, the paper's contributions and critical findings are threefold:

- 1) A cross-level strategy for load smoothing is proposed for IPs, which integrates load forecasting, dispatch, and execution, to optimize system efficiency, resulting in improved energy efficiency utilization.
- 2) Develop a load forecasting utilization method. The elasticity of heat network is considered as a buffer device to improve the stability of steam supply. By transforming short-term demand predictions into production-side energy supply predictions, system performance is improved.
- 3) Production smoothness is prioritized. By introducing constraints for minimal changes in equipment load and operational parameters, the stability of equipment operation and the efficiency of load regulation models are improved.

While this paper considers the elasticity of the steam pipeline network, it does not delve deeply into its mechanism. Future work may involve further integration of the mechanism model of the steam pipeline network. Additionally, it is important to explore more convenient methods of adaptive updating of the model, particularly for online applications.

CRedit authorship contribution statement

Xiaojie Lin: Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Xueru Lin:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Wei Zhong:** Writing – review & editing, Supervision, Conceptualization. **Feiyun Cong:** Writing – review & editing, Supervision, Conceptualization. **Yi Zhou:** Validation, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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References

- [1] Yang W, Zhang E, Zhao J, et al. Dawn of clean energy: Enhanced heat transfer, radiative cooling, and firecracker-style controlled nuclear fusion power generation system[J]. *Clean Energy Science and Technology* 2023;1(1):61.
- [2] Parimi K, Kaur B, Lankapalli S, et al. Performance, combustion, and emission characteristics of a diesel engine fuelled with hydrogen compressed natural gas and Kusum seed biodiesel[J]. *Waste Disposal & Sustainable Energy* 2023;1–13.
- [3] Wang Y, Ren H, Dong L, et al. Smart solutions shape for sustainable low-carbon future: A review on smart cities and industrial parks in China[J]. *Technological Forecasting and Social Change* 2019;144:103–17.
- [4] Xue R, Wang S, Long W, et al. Uncovering GHG emission characteristics of industrial parks in Central China via emission inventory and cluster analysis[J]. *Energy Policy* 2021;151(7):112191.
- [5] Lyu Y, Liu Y, Guo Y, et al. Review of green development of Chinese industrial parks [J]. *Energy Strategy Reviews* 2022;42:100867.
- [6] Yu X, Zheng H, Sun L, et al. An emissions accounting framework for industrial parks in China[J]. *Journal of Cleaner Production* 2019;244(3):118712.
- [7] Butturi M, Lolli F, Sellitto M, et al. Renewable energy in eco-industrial parks and urban-industrial symbiosis: A literature review and a conceptual synthesis[J]. *Applied Energy* 2019;255:113825.
- [8] Sharma S. Energy management in textile industry[J]. *International Journal of Power System Operation and Energy Management* 2012;2(1):2.
- [9] Khude P. A review on energy management in textile industry[J]. *Innovative Energy & Research* 2017;6(169):2.
- [10] Ozturk E, Karaboyaci M, Yetis U, et al. Evaluation of integrated pollution prevention control in a textile fiber production and dyeing mill[J]. *Journal of cleaner production* 2015;88:116–24.
- [11] Bank L, Rösch M, Unterberger E, et al. Comparison of simulation-based and optimization-based energy flexible production planning[J]. *Procedia CIRP* 2019; 81:294–9.
- [12] Boix M, Montastruc L, Pibouleau L, et al. Industrial water management by multiobjective optimization: from individual to collective solution through eco-industrial parks[J]. *Journal of Cleaner Production* 2012;22(1):85–97.
- [13] Boix M, Montastruc L, Azzaro-Pantel C, et al. Optimization methods applied to the design of eco-industrial parks: a literature review[J]. *Journal of Cleaner Production* 2015;87:303–17.
- [14] Cao Y, Li D, Zhang Y, et al. Optimal energy management for multi-microgrid under a transactive energy framework with distributionally robust optimization[J]. *IEEE Transactions on Smart Grid* 2021;13(1):599–612.
- [15] Lü X, Wu Y, Lian J, et al. Energy management of hybrid electric vehicles: A review of energy optimization of fuel cell hybrid power system based on genetic algorithm [J]. *Energy Conversion and Management* 2020;205:112474.
- [16] Ocampo E, Huang Y, Kuo C. Feasible reserve in day-ahead unit commitment using scenario-based optimization[J]. *Energies* 2020;13(20):5239.

- [17] Ponciroli R, Stauff N, Ramsey J, et al. An improved genetic algorithm approach to the unit commitment/economic dispatch problem[J]. *IEEE Transactions on Power Systems* 2020;35(5):4005–13.
- [18] Zhai Y, Mu N, Liao X, et al. Unit commitment problem using an efficient pso based algorithm[C]. // In: 2019 Eleventh International Conference on Advanced Computational Intelligence (ICACI). IEEE; 2019. p. 320–4.
- [19] Tiwari S, Dwivedi S, Dave M. A multi-stage hybrid artificial intelligence based optimal solution for energy storage integrated mixed generation unit commitment problem[J]. *Journal of Intelligent & Fuzzy Systems* 2018;35(5):4909–19.
- [20] Zhu X, Yang J, Pan X, et al. Regional integrated energy system energy management in an industrial park considering energy stepped utilization[J]. *Energy* 2020;201:117589.
- [21] Zhou X, Zhang H, Fang S, et al. Off-design performance analysis with various operation methods for ORC-based compression heat recovery system in cryogenic air separation units[J]. *Energy* 2022;261:125364.
- [22] Jia J, Lv Y. Operation optimization of CFB boiler based on data-driven and pattern mining[J]. *Thermal Power Generation* 2020;49(5):33–9.
- [23] Wang C, Liu Y, Zheng S, et al. Optimizing combustion of coal fired boilers for reducing NOx emission using Gaussian Process[J]. *Energy* 2018;153:149–58.
- [24] Kang J, Niu Y, Zhang G, et al. Study on case based reasoning adaptive optimization method and its application in power plant boiler combustion system[J]. *Chinese Journal of Scientific Instrument* 2019;40(12):214–23.
- [25] Tang Z, Zhang Z. The multi-objective optimization of combustion system operations based on deep data-driven models[J]. *Energy* 2019;182:37–47.
- [26] Blackburn L, Tuttle J, Andersson K, et al. Development of novel dynamic machine learning-based optimization of a coal-fired power plant[J]. *Computers & Chemical Engineering* 2022;163:107848.
- [27] Wu Q, Ren H, Shi S, et al. Analysis and prediction of industrial energy consumption behavior based on big data and artificial intelligence[J]. *Energy Reports* 2023;9:395–402.
- [28] Eldabah M, Nassar I, Esmail A. Enhancement of electrical distribution networks performance using the load management methodology[J]. *Energy Reports* 2020;6 (November 2020):2066–74.
- [29] Kerckänen A, Korpela J, Huiskonen J. Demand forecasting errors in industrial context: Measurement and impacts[J]. *International Journal of Production Economics* 2009;118(1):43–8.
- [30] Li J, Herdem M, Nathwani J, et al. Methods and applications for Artificial Intelligence, Big Data, Internet of Things, and Blockchain in smart energy management[J]. *Energy and AI* 2023;11:100208.
- [31] Ahmad T, Zhang H, Yan B. A review on renewable energy and electricity requirement forecasting models for smart grid and buildings[J]. *Sustainable Cities and Society* 2020;55:102052.
- [32] Alkhayat G, Mehmood R. A review and taxonomy of wind and solar energy forecasting methods based on deep learning[J]. *Energy and AI* 2021;4:100060.
- [33] Gaffar A, Aisjah A. Load forecasting for power system planning and operation using artificial neural network (a case study on Larona Hydro Power in the nickel smelting plant)[C]. // In: AIP Conference Proceedings. 2088. AIP Publishing LLC; 2019. 030016.
- [34] Zhang Y, Deng C, Zhao R, et al. A novel integrated price and load forecasting method in smart grid environment based on multi-level structure[J]. *Engineering Applications of Artificial Intelligence* 2020;95:103852.
- [35] Yin L, Xie J. Multi-temporal-spatial-scale temporal convolution network for short-term load forecasting of power systems[J]. *Applied Energy* 2021;283:116328.
- [36] Wang S, Wang S, Chen H, et al. Multi-energy load forecasting for regional integrated energy systems considering temporal dynamic and coupling characteristics[J]. *Energy* 2020;195:116964.
- [37] Ribeiro G, Mariani V, Coelho L. Enhanced ensemble structures using wavelet neural networks applied to short-term load forecasting[J]. *Engineering Applications of Artificial Intelligence* 2019;82(JUN):272–81.
- [38] Liu J, Chen J, Yan G, et al. Clustering and dynamic recognition based auto-reservoir neural network: A wait-and-see approach for short-term park power load forecasting[J]. *Isience* 2023;26(8).
- [39] You M, Wang Q, Sun H, et al. Digital twins based day-ahead integrated energy system scheduling under load and renewable energy uncertainties[J]. *Applied Energy* 2022;305:117899.
- [40] Zhang L, Shi J, Wang L, et al. Electricity, heat, and gas load forecasting based on deep multitask learning in industrial-park integrated energy system[J]. *Entropy* 2020;22(12):1355.
- [41] Chen B, Wang Y. Short-term electric load forecasting of integrated energy system considering nonlinear synergy between different loads[J]. *IEEE Access* 2021;9:43562–73.
- [42] Zhao H, Guo S. Uncertain interval forecasting for combined electricity-heat-cooling-gas loads in the integrated energy system based on multi-task learning and multi-kernel extreme learning machine[J]. *Mathematics* 2021;9(14):1645.
- [43] Kuang W, Yang H, Ying C, et al. Cost-effective, environmentally-sustainable and scale-up synthesis of vertically oriented graphenes from waste oil and its supercapacitor applications[J]. *Waste Disposal & Sustainable Energy* 2021;3:31–9.
- [44] Shaqsi A, Sopian K, Al-Hinai A. Review of energy storage services, applications, limitations, and benefits[J]. *Energy Reports* 2020;6:288–306.
- [45] Mirzaei M, Nazari-Heris M, Mohammadi-Ivatloo B, et al. Robust flexible unit commitment in network-constrained multicarrier energy systems[J]. *IEEE Systems Journal* 2020;15(4):5267–76.
- [46] Hemmati R. Optimal cogeneration and scheduling of hybrid hydro-thermal-wind-solar system incorporating energy storage systems[J]. *Journal of Renewable and Sustainable Energy* 2018;10(1).
- [47] Hussain A, Musilek P. Energy management of buildings with energy storage and solar photovoltaic: A diversity in experience approach for deep reinforcement learning agents[J]. *Energy and AI* 2024;15:100313.
- [48] Boix M, Négny S, Montastruc L, et al. Flexible networks to promote the development of industrial symbioses: A new optimization procedure[J]. *Computers & Chemical Engineering* 2023;169:108082.
- [49] Huang W, Zhang X, Li K, et al. Resilience oriented planning of urban multi-energy systems with generalized energy storage sources[J]. *IEEE Transactions on Power Systems* 2021;37(4):2906–18.
- [50] Chen C, Wu X, Li Y, et al. Distributionally robust day-ahead scheduling of park-level integrated energy system considering generalized energy storages[J]. *Applied Energy* 2021;302:117493.
- [51] Hao H, Wu D, Lian J, et al. Optimal coordination of building loads and energy storage for power grid and end user services[J]. *IEEE Transactions on Smart Grid* 2017;9(5):4335–45.
- [52] Li X, Liu J, Wang K, et al. Performance analysis of reheat steam temperature control system of thermal power unit based on constrained predictive control[J]. *Complexity* 2019;2019(11):1–12.
- [53] Lin X, Wu Y, Lin X, et al. Real-Time Optimization of Power Plant Boiler Based on Pattern Matching and Neural Networks[C]. // In: 2022 International Conference on High Performance Big Data and Intelligent Systems (HDIS). IEEE; 2022. p. 282–6.